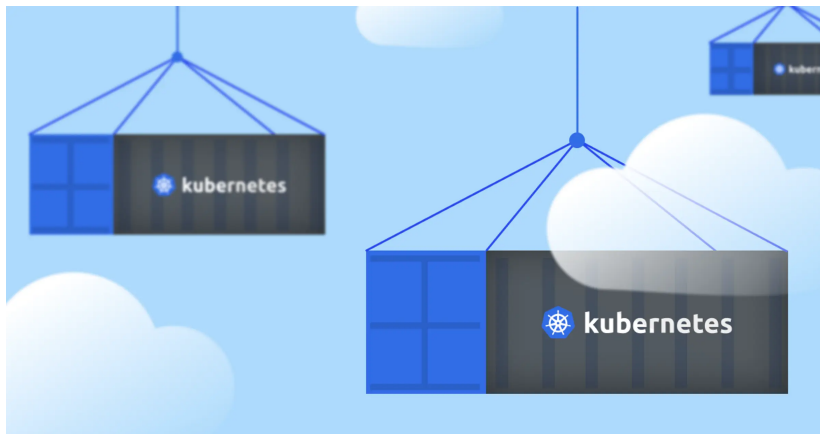




Custom k8s deployed on Azure

1 Reasons

- List of Reasons

2 Fundamentals

3 Costs

4 Customization

5 Scaling

6 Summary

Reasons for Considering Alternative(s)



Figure: Cost !!!

- The cost of an AKS cluster Super High!
- Azure Kubernetes Service (AKS)
 - k8s Node(s)

Reasons for Considering Alternative(s)

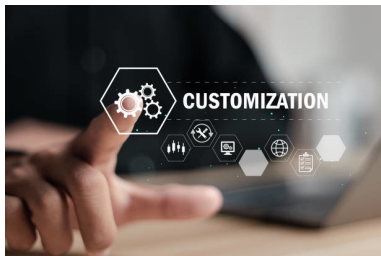


Figure: Customization

- The cost of an AKS cluster **Super High!**
- **Azure Kubernetes Service (AKS)**
 - k8s Node(s)
 - No SSH to Node(s)
 - No Options for OS
 - One choice for all
 - Ability to split the node(s) to different region(s)

Reasons for Considering Alternative(s)

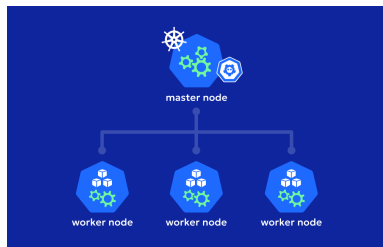


Figure: k8s Cluster

- The cost of an AKS cluster **Super High!**
- Azure Kubernetes Service (**AKS**)
 - **k8s Node(s)**
 - No SSH to Node(s)
 - No Options for OS
 - One choice for all
 - Ability to split the node(s) to different region(s)

Reasons for Considering Alternative(s)



Figure: Troubleshooting

- The cost of an AKS cluster **Super High!**
- Azure Kubernetes Service (**AKS**)
 - k8s Node(s)
 - **No SSH to Node(s)**
 - No Options for OS
 - One choice for all
 - Ability to split the node(s) to different region(s)

Reasons for Considering Alternative(s)



Figure: Operating System

- The cost of an AKS cluster **Super High!**
- Azure Kubernetes Service (**AKS**)
 - k8s Node(s)
 - No SSH to Node(s)
 - **No Options for OS**
 - One choice for all
 - Ability to split the node(s) to different region(s)

Reasons for Considering Alternative(s)

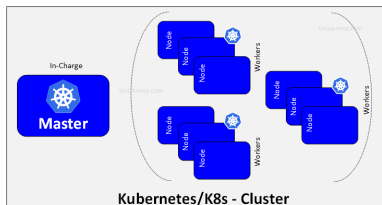


Figure: Node(s) Scaling

- The cost of an AKS cluster **Super High!**
- Azure Kubernetes Service (**AKS**)
 - k8s Node(s)
 - No SSH to Node(s)
 - No Options for OS
 - **One choice for all**
 - Ability to split the node(s) to different region(s)

Reasons for Considering Alternative(s)

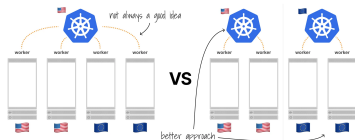


Figure: k8s Region(s)

- The cost of an AKS cluster **Super High!**
- Azure Kubernetes Service (**AKS**)
 - k8s Node(s)
 - No SSH to Node(s)
 - No Options for OS
 - One choice for all
 - **Ability to split the node(s) to different region(s)**

1 *Reasons*

2 *Fundamentals*

- HLD Local ETCD
- HLD External ETCD

3 *Costs*

4 *Customization*

5 *Scaling*

6 *Summary*

Local ETCD

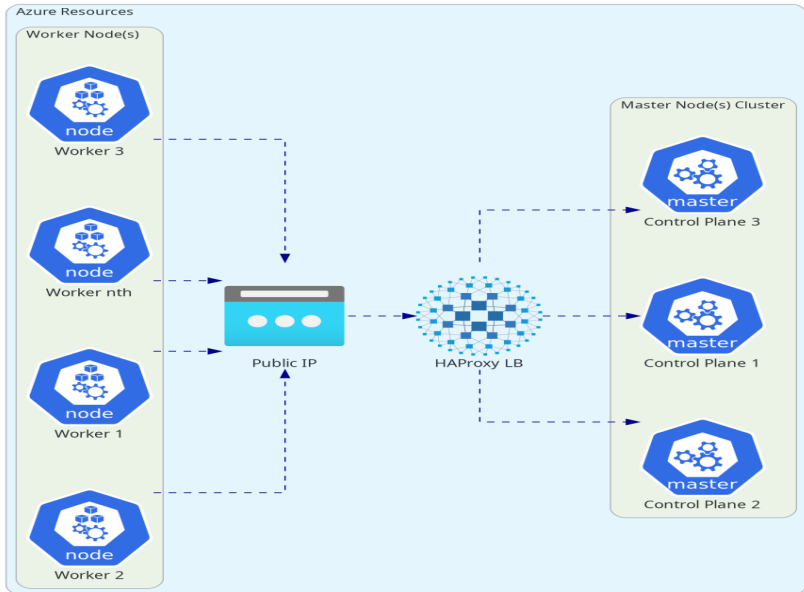


Figure: HLD Local ETCD

1 *Reasons*

2 *Fundamentals*

- HLD Local ETCD
- HLD External ETCD

3 *Costs*

4 *Customization*

5 *Scaling*

6 *Summary*

External ETCD

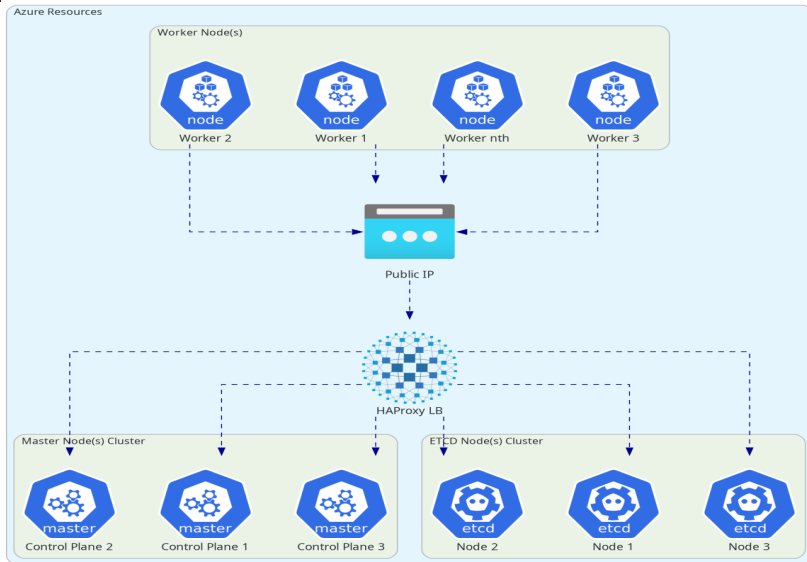


Figure: HLD External ETCD

- Custom k8s Costs

1 Reasons

2 Fundamentals

3 **Costs**

- AKS Costs

4 Customization

5 Scaling

6 Summary

AKS Costs

How Expensive is AKS?

Your Estimate

Your Estimate

Azure Kubernetes Service (AKS) Standard: Cluster management for 1 clusters; 1 B2sv2-series v2; 2 vCPUs, 4 GB RAM, 0 GB Temporary storage, DKK 0.292/hour... Upfront: DKK 0.00 Monthly: DKK 768.34

Azure Kubernetes Service (AKS)

Region: North Europe Tier: Standard

Cluster Management

Standard tier is recommended for all production workload, large and mission-critical applications. Standard tier charges on a per cluster basis and comes with a guaranteed SLA, scalable control plane, and higher node limits per cluster.

1 Cluster DKK 522.57

Nodes

Operating system: Linux

Category: All Instance Series: B2sv2-series

INSTANCE: [Need help finding the right VM?](#) B2sv2 v2: 2 vCPUs, 4 GB RAM, 0 GB Temporary storage, DKK 0.292/hour... 1 x 1 Month

Virtual Machines

Figure: Basic 770 DKK / month !!!

How Expensive is AKS?

	Free tier	Standard tier	Premium tier
When to use	• You want to experiment with AKS at no extra cost • You're new to AKS and Kubernetes	• You're running production or mission-critical workloads and need high availability and reliability • You need a financially backed SLA • Automatically selected for AKS automatic clusters	• You're running production or mission-critical workloads and need high availability and reliability • You need a financially backed SLA • All mission critical, at scale, or production workloads requiring two years of one Kubernetes version support
Supported cluster types	• Development clusters or small scale testing environments • Clusters with fewer than 10 nodes	• Enterprise-grade or production workloads • Clusters with up to 5,000 nodes	• Enterprise-grade or production workloads • Clusters with up to 5,000 nodes
Pricing	• Free cluster management • Pay-as-you-go for resources you consume	• Pay-as-you-go for resources you consume • Standard tier Cluster Management Pricing ☑	• Pay-as-you-go for resources you consume • Premium tier Cluster Management Pricing ☑
Feature comparison	• Recommended for clusters with fewer than 10 nodes, but can support up to 1,000 nodes • Includes all current AKS features	• Uptime SLA is enabled by default • Greater cluster reliability and resources • Can support up to 5,000 nodes in a cluster • Includes all current AKS features	• Includes all current AKS features from standard tier • Microsoft maintenance past community support

Figure: Basic and Premium

How Expensive is AKS?

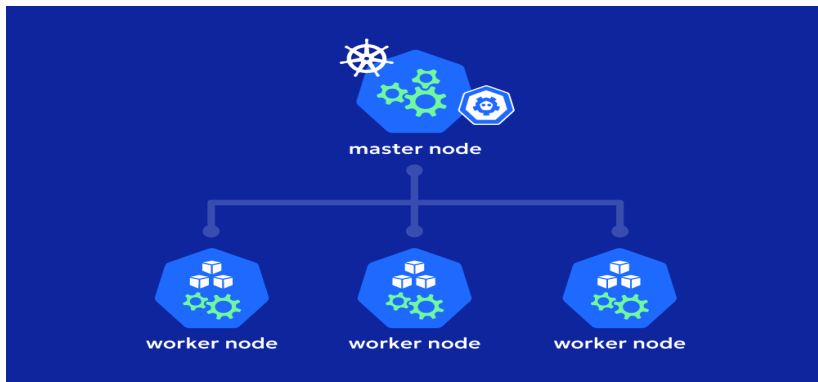


Figure: Basic is not Really HA !!!

AKS Costs

How Expensive is AKS?

The screenshot shows the Azure Cost Estimator for the Azure Kubernetes Service (AKS) Premium tier. The configuration includes:

- Region:** North Europe
- Tier:** Premium
- Cluster Management:** Premium tier (recommended for mission critical workloads).
- Nodes:** Linux operating system, General purpose category, Av2 Standard instance series.
- Instance Size:** A2 v2: 2 Cores, 4 GB RAM, 20 GB Temporary storage, DKK 0.616/hour.
- Virtual Machines:** 3 VMs, 1 Month.

The total estimated cost is **DKK 4,491.67** Monthly. A breakdown shows the Cluster Management cost at DKK 3,135.42.

Figure: Premium 4000 DKK / Month !!!

How Expensive is AKS?

Support

SUPPORT:
Basic (Included) ⓘ

Select your program/offer

LICENSING PROGRAM:
Microsoft Customer Agreement (MCA) ⓘ

Show Dev/Test Pricing ⓘ

Estimated upfront cost

Estimated monthly cost

Basic (Included)
Basic support is included for companies or individuals using Azure in a non-production environment or for trial and evaluation. [Learn more](#)

Developer
Developer support is for companies or individuals using Azure in a non-production environment or for trial and evaluation. [Learn more](#)

Standard
Standard support is for small- or mid-size companies with minimal business-critical dependence on Azure. [Learn more](#)

Professional Direct
Professional Direct support is for mid-size to large companies with substantial business-critical use of Azure. [Learn more](#)

DKK 0.00

DKK 0.00

DKK 1,160.65

CURRENCY
Denmark - Krone (kr) DKK
1 USD = 7.16 DKK

To get the most accurate pricing estimate, [sign in](#) to your account and ensure you've selected the correct licensing program.

Prices are estimates only and are not intended as actual prices. Prices are calculated based on the last business day of the previous month end. If the two business days prior to the end of the month fall on a bank holiday in major markets, the rate setting day is generally the day immediately preceding the two business days. This rate applies to all transactions during the upcoming month. Sign in to the [Azure pricing calculator](#) to see pricing based on your current program. [Contact us](#) or [view our Azure sales calculator](#) for more information on pricing as it relates to your Azure usage. For forecasts, see [Azure pricing calculator](#).

Export Save Share Log in to save

Chat with Sales

Figure: Support level(s)

How Expensive is AKS?

Support

SUPPORT: Professional Direct DKK 7,158.50

Select your program/offer

LICENSING PROGRAM: Microsoft Customer Agreement (MCA) Log in to see your Azure agreement pricing.

Show Dev/Test Pricing

Estimated upfront cost DKK 0.00

Estimated monthly cost DKK 8,319.15

Export Save Share Log in to save and share cost estimates.

CURRENCY: Denmark - Krone (dkr) DKK
1 USD = 7.16 DKK

To get the most accurate pricing estimate, sign in to your account and ensure you've selected the correct licensing program.

Prices are estimates only and are not intended as actual price quotes. Actual pricing may vary depending on the type of agreement entered with Microsoft, date of purchase, and the currency exchange rate. Prices are calculated based on US dollars and converted using London closing spot rates that are captured in the two business days prior to the last business day of the previous month end. If the two business days prior to the end of the month fall on a bank holiday in major markets, the rate setting day is generally the day immediately preceding the two business days. This rate applies to all transactions during the upcoming month. Sign in to the [Azure pricing calculator](#) to see pricing based on your current account setup with Microsoft. Contact an [Azure sales specialist](#) for more information on pricing or to request a price quote. For frequently asked questions, see [pricing FAQ](#).

Chat with Sales

Figure: Cost of Support Premium level 7160 DKK !!!

● Custom k8s Costs

1 Reasons

2 Fundamentals

3 **Costs**
● AKS Costs

4 Customization

5 Scaling

6 Summary

Custom k8s Costs

How Expensive is custom k8s?

Your Estimate

Your Estimate

Virtual Machines 1 B2als v2 (2 vCPUs, 4 GB RAM) x 1 Month (Pay as y... Upfront: DKK 0.00 Monthly: DKK 213.21

Virtual Machines

Get DKK 200 credit plus free monthly amounts of popular services for 12 months—including Virtual Machines. [See free amounts](#)

Region: North Europe Operating system: Linux Type: Ubuntu Tier: Standard

Category: All Instance Series: Basv2-series INSTANCE: [\(Need help finding the right VM?\)](#) B2als v2: 2 vCPUs, 4 GB RAM, 0 GB Temporary storage, DKK 0.292/hour

1 x 1 Month Virtual machines

Chat with Sales

Figure: Basic 214 DKK / month (including SSD disc) !!!

1 *Reasons*

2 *Fundamentals*

3 *Costs*

4 *Customization*
● Node customization

5 *Scaling*

6 *Summary*

Node customization

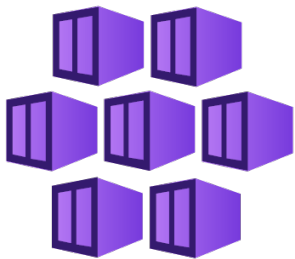


Figure: AKS

- No customization with AKS
- Custom Linux VM 100% Customization
 - RedHat OS
 - Core OS
 - Debian OS
 - Oracle Linux OS
 - SUSE Linux EE OS
 - SUSE Linux OS
 - Ubuntu OS

Node customization



Figure: Azure

- No customization with AKS
- Custom Linux VM
100% Customization
 - RedHat OS
 - Core OS
 - Debian OS
 - Oracle Linux OS
 - Suse Linux EE OS
 - Suse Linux OS
 - Ubuntu OS



Figure: RedHat OS

- No customization with AKS
- Custom Linux VM 100% Customization
 - RedHat OS
 - Core OS
 - Debian OS
 - Oracle Linux OS
 - Suse Linux EE OS
 - Suse Linux OS
 - Ubuntu OS



Figure: Core OS

- No customization with AKS
- Custom Linux VM 100% Customization
 - RedHat OS
 - **Core OS**
 - Debian OS
 - Oracle Linux OS
 - Suse Linux EE OS
 - Suse Linux OS
 - Ubuntu OS



Figure: Debian OS

- No customization with AKS
- Custom Linux VM 100% Customization
 - RedHat OS
 - Core OS
 - **Debian OS**
 - Oracle Linux OS
 - Suse Linux EE OS
 - Suse Linux OS
 - Ubuntu OS



Figure: Oracle Linux OS

- No customization with AKS
- Custom Linux VM 100% Customization
 - RedHat OS
 - Core OS
 - Debian OS
 - **Oracle Linux OS**
 - Suse Linux EE OS
 - Suse Linux OS
 - Ubuntu OS



Figure: Suse EE OS

- No customization with AKS
- Custom Linux VM 100% Customization
 - RedHat OS
 - Core OS
 - Debian OS
 - Oracle Linux OS
 - Suse Linux EE OS
 - Suse Linux OS
 - Ubuntu OS



Figure: Suse OS

- No customization with AKS
- Custom Linux VM 100% Customization
 - RedHat OS
 - Core OS
 - Debian OS
 - Oracle Linux OS
 - Suse Linux EE OS
 - Suse Linux OS
 - Ubuntu OS



Figure: Ubuntu OS

- No customization with AKS
- Custom Linux VM 100% Customization
 - RedHat OS
 - Core OS
 - Debian OS
 - Oracle Linux OS
 - Suse Linux EE OS
 - Suse Linux OS
 - **Ubuntu OS**

Node(s) Scaling

1 *Reasons*

2 *Fundamentals*

3 *Costs*

4 *Customization*

5 *Scaling*
● Node(s) Scaling

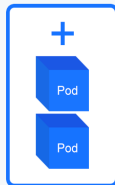
6 *Summary*

Node(s) Scaling

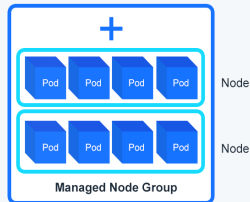
Different Methods for Autoscaling in **Kubernetes**



**Horizontal Pod
Autoscaler (HPA)**



**Vertical Pod
Autoscaler (VPA)**



**Cluster
Autoscaler**

Figure: Horizontal and Vertical Scaling

Node(s) Scaling

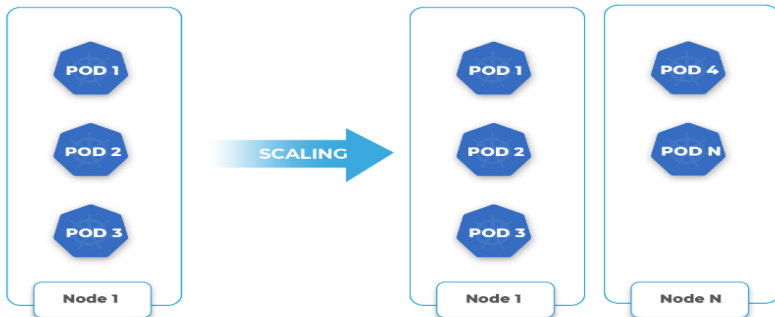


Figure: Scaling with Custom k8s VS AKS

Key points repetition

1 *Reasons*

2 *Fundamentals*

3 *Costs*

4 *Customization*

5 *Scaling*

6 *Summary*
● Key points repetition

Summary

- In section “Reasons” page “1” Listed the reasons!
- In section “Fundamentals” page “2” Kubernetes cluster fundamentals!
- In section “Costs” page “3” Comparison of costs (per version)!
- In section “Customization” page “4” How customizable is each one!
- In section “Scaling” page “5” Discussed the scaling for different clusters!

Summary

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Coming up: Q & A